

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:30-12-24**
Time: 3HRS **GTM-6** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

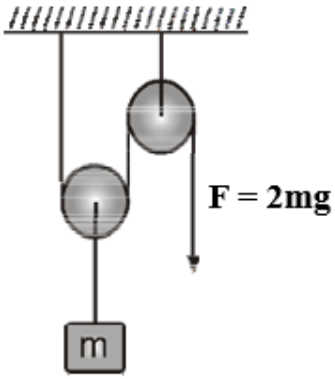
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In the relation $P=\frac{\alpha}{\beta}e^{-\frac{\alpha Z}{k\theta}}$; P is pressure, Z is distance, k is Boltzmann constant and θ is the temperature. The dimensions of β will be:

   A) $[M^0L^2T^0]$ B) $[ML^2T]$ C) $[ML^0T^{-1}]$ D) $[M^0L^2T^{-1}]$

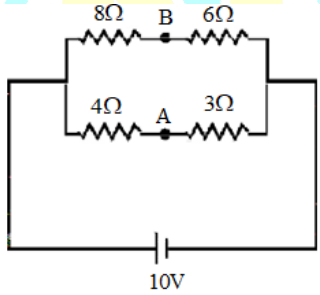
2. In the shown mass pulley system, pulleys and strings are massless. The one end of the string is pulled by the force F = 2mg. The acceleration of the block will be x m/s^2. Then the value of x will be ($g = 10\ m/s^2$)

For More Material Join: @JEEAdvanced_2025



A) 20 B) 30 C) 40 D) 50

3. The potential difference between point A and B is



A) $\frac{20}{7}V$ B) $\frac{40}{7}V$ C) $\frac{10}{7}V$ D) 0

4. A wire of cross-sectional area A is stretched horizontally between two clamps located at a distance $2l$ meters from each other. A weight W is suspended from the midpoint of the wire. The strain produced in the wire, (if the vertical distance through which the midpoint of the wire moves down by x where $x << l$) will be

A) $\frac{x^2}{l^2}$ B) $\frac{2x^2}{l^2}$ C) $\frac{x^2}{2l^2}$ D) $\frac{x}{2l}$

5. A metal surface is illuminated by light of two different wavelengths 248 nm and 310nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are μ_1 and μ_2, respectively. If the ratio $\mu_1 : \mu_2 = 2:1$ and hc = 1240 eV nm, the work function of the metal is nearly ----------

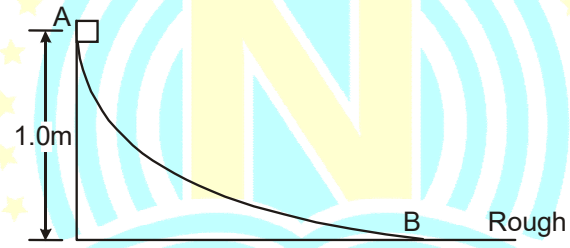
A) 4.3 eV B) 3.7 eV C) 1.7eV D) 2.7 eV

For More Material Join: @JEEAdvanced_2025

6. Consider a gas of hydrogen like ions in an excited state A. It emits photons having wavelength equal to the wavelength of the first line of the Lyman series together with photons of five other wavelengths. Find the principal quantum number of the state A.

A) 4 B) 2 C) 5 D) 6

7. A block weighing 10 N travels down a smooth curved track AB joined to a rough horizontal surface (figure). The rough surface has a friction coefficient of 0.20 with the block. If the block starts slipping on the track from a point 1.0 m above the horizontal surface, the distance it will move on the rough surface is ___ m.



A) 25 B) 50 C) 30 D) 5

8. The electric field part of an electromagnetic wave in a medium is represented by

$E_x = 0;\quad E_y = 2.5\frac{N}{C}\cos\left[\left(2\pi\times10^{6}\frac{rad}{s}\right)t-\left(\pi\times10^{-2}\frac{rad}{m}\right)x\right];\ E_z = 0$. The wave is :

A) moving along y direction with frequency 10^6 Hz and wavelength 200m.

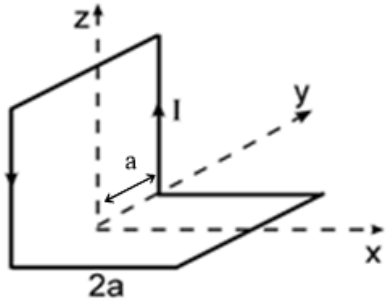
B) moving along x direction with frequency 10^6 Hz and wavelength 100m

C) moving along x direction with frequency 10^6 Hz and wavelength 200m

D) moving along –x direction with frequency 10^6 Hz and wavelength 200m

9. A non-planer loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P(a, 0, a) points in the direction of …...

For More Material Join: @JEEAdvanced_2025



A) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$ B) $\frac{1}{\sqrt{3}}\left(-\hat{j}+\hat{k}+\hat{i}\right)$ C) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$ D) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$

10. A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60cm. The aperture is illuminated normally by a parallel beam of wavelength 5×10^{-5} cm. The distance of the first dark band of the diffraction pattern from the centre of the screen is :

A) 0.15 cm B) 0.10 cm C) 0.25 cm D) 0.20 cm

11. A transformer is used for a 100 watt, 20 volt electric bulb at a place where the A.C. mains potential is 200 volt and the current drawn is 0.6 A. The efficiency of the transformer is nearly

A) 83.33% B) 73% C) 50% D) 62%

12. A point P lies on the axis of a fixed ring of mass M and radius R, at a distance 2R from its centre O. A small particle starts from P and reaches O under gravitational attraction only. Its speed at O will be

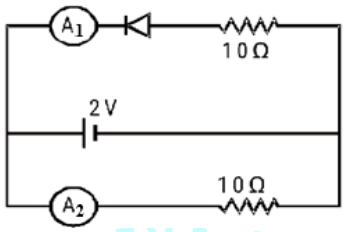
A) zero B) $\sqrt{\frac{2GM}{R}}$ C) $\sqrt{\frac{2GM}{R}\left(\sqrt{5}-1\right)}$ D) $\sqrt{\frac{2GM}{R}\left(1-\frac{1}{\sqrt{5}}\right)}$

13. A platinum wire has resistance of 10 ohm at 0ºC and 20 ohm at 273ºC. Find the value of temperature coefficient of resistance.

A) $\frac{1}{273}\,{}^{0}C^{-1}$ B) $\frac{3}{273}\,{}^{0}C^{-1}$ C) $\frac{5}{273}\,{}^{0}C^{-1}$ D) $\frac{7}{273}\,{}^{0}C^{-1}$

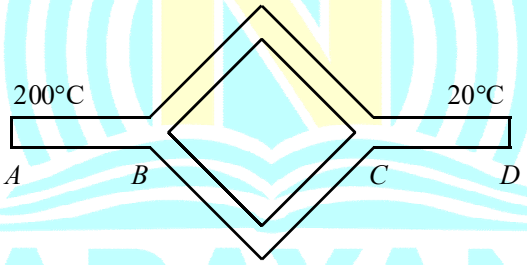
For More Material Join: @JEEAdvanced_2025

14. In the following circuit readings in ammeters A_1 and A_2 will be _____



A) 0.2 A, zero B) zero, 0.2 A C) 0.2A, 0.2A D) 0.2A, 0.4A

15. Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperatures $200°C$ and $20°C$, respectively. The temperature of junction B will be



A) $120°C$ B) $100°C$ C) $140°C$ D) $80°C$

16. Find out the electric potential due to a dipole of dipole moment $\vec{P} = \hat{i} + \sqrt{3}\,\hat{j}$ kept at origin at point $\left(-1, \sqrt{3}, 0\right)$. [Take $k = \frac{1}{4\pi \in_0}$]

A) $V = \frac{K}{4}$ B) $V = \frac{3K}{4}$ C) $V = K$ D) $V = \frac{5K}{4}$

17. 250 g ball strikes at the bat with velocity of 10 m/s and remains in contact up to 0.01 second after then it returns with same speed then find out force exerted by ball on bat:

A) 25 N B) 50 N C) 250 N D) 500 N

18. A particle of mass m is projected with a velocity v making an angle 45° with the horizontal. The magnitude of the angular momentum of the projectile about the point of projection when the particle is at its maximum height h, is:-

A) Zero B) $\frac{mv^3}{4\sqrt{2}\,g}$ C) $\frac{mv^3}{16\sqrt{2}\,g}$ D) $\frac{mv^3}{3\sqrt{2}\,g}$

For More Material Join: @JEEAdvanced_2025

19. A gas is filled in a rigid container at pressure P_0. If the mass of each molecule is halved keeping the total number of molecules same and their r.m.s. speed is doubled then find the new pressure.

A) $2P_0$ B) P_0 C) $3P_0$ D) $5P_0$

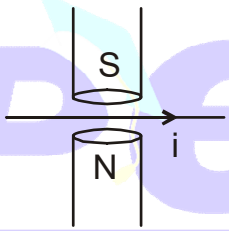
20. The self-inductance of a choke coil is 10 mH. When it is connected with a 10V D.C. source, then the loss of power is 20 watt. When it is connected with 10 volt A.C. source loss of power is 10 watt. The frequency of A.C. source will be 'x' Hz, the value of x is (nearly)

A) 30Hz B) 60Hz C) 80Hz D) 100Hz

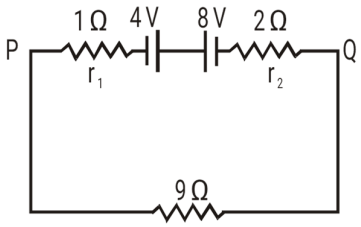
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A magnetic field of strength 1.0 T is produced by a strong electromagnet in a cylindrical region of diameter 4.0 cm as shown in figure. A wire, carrying a current of 2.0 A, is placed perpendicular to the field and intersecting the axis of the cylindrical region. Find the magnitude of the force acting on the wire is _____ $\times 10^{-2}$ N



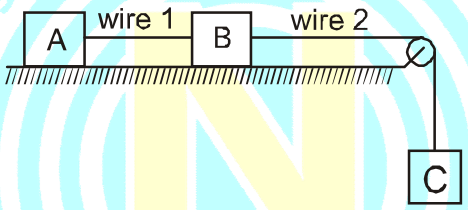
22. Two batteries of e.m.f. 4 V and 8 V with internal resistances 1Ω (r_1) and 2Ω (r_2) are connected in a circuit with a resistance of 9Ω as shown in figure. The potential difference between the points P and Q is _____ volts.



For More Material Join: @JEEAdvanced_2025

23. In a hydrogen like atom, electron makes transition from an energy level with quantum number n to another with quantum number $(n-1)$. If $n >> 1$, the frequency of radiation emitted is proportional to n^{-x}. Find the value of x.

24. Three blocks A, B and C each of mass 4 kg are attached as shown in figure. Both the wires have equal cross sectional area 5×10^{-7} m^2. The surface is smooth. If the longitudinal strain in wire 2 is $\frac{x}{3}\times10^{-4}$, then the x will be (if Young modulus of both the wires is 2×10^{11} N/m^2) (take $g = 10$ m/s^2)



25. A luminous point object is moving along the principal axis of a concave mirror of focal length 12 cm towards it. When its distance from the mirror is 20 cm its velocity is 4 cm/s. The speed of the image in cm/s at that instant is

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Given below are two statements:

Statement-I: The colourless, odourless gas is evolved with brisk effervescence on warming a salt with dil. H_2SO_4, turns a lime water milky, it is a identification test for CO_3^{2-}.

Statement-II: In statement-I the milky white is due to formation of calcium carbide.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) Both Statement-I and Statement-II are false

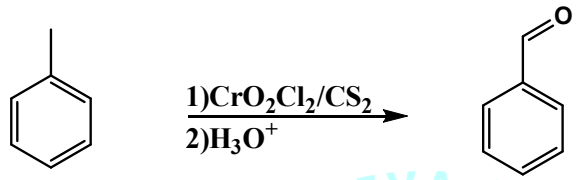
B) Statement-I is false but Statement-II is true

For More Material Join: @JEEAdvanced_2025

C) Statement-I is true but Statement-II is false

D) Both Statement-I and Statement-II are true

27.

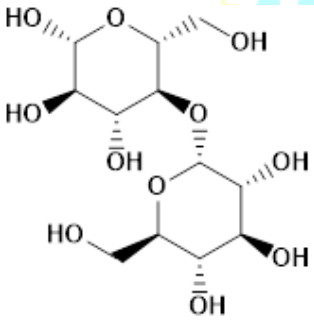


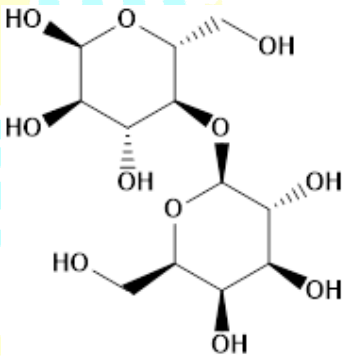
This reaction is known as

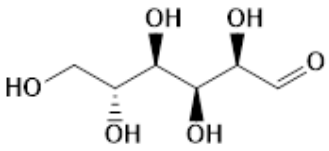
A) Friedel crafts reaction B) Perkin reaction

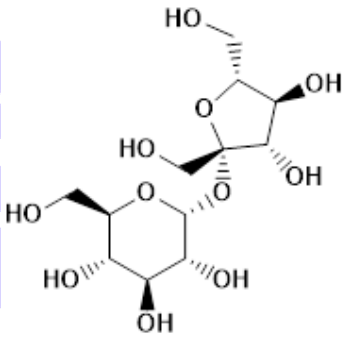
C) Gattermann reaction D) Etard reaction

28. Sugar which does not give silver mirror with tollens reagent is:

A) 

B) 

C) 

D) 

29. Given below are the two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): There is a considerable increase in covalent radius from O to S. However from Se to Po only a small increase in covalent radius is observed.

For More Material Join: @JEEAdvanced_2025

Reason (R): In 16th group covalent and ionic radii in a particular oxidation state of an element increases down the group.

In the light of the above statement, choose the most appropriate answer from the options given below:

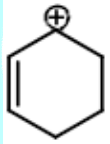
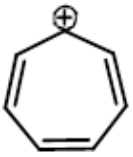
A) (A) is false but (R) is true

B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

C) (A) is true but (R) is false

D) Both (A) and (R) are true and (R) is the correct explanation of (A)

30. Which of the following molecule/species is most stable?

A)  B)  C)  D) 

31. Diamagnetic actinoid ions are

A) Am^{+3} and Cm^{+3} B) Es^{+3} and Fm^{+3} C) Ac^{+3} and Lr^{+3} D) Np^{+3} and Pu^{+3}

32. Identify the correct electronegativity order

A) B > Tl > In > Ga > Al B) B > Al > Tl > In > Ga

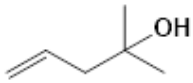
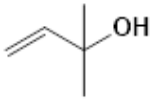
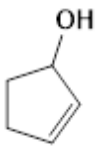
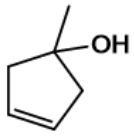
C) B > Al > Ga > Tl > In D) B > In > Ga > Tl > Al

33. What is the wavelength (in nm) of a photon emitted during a transition from n = 5 state to n = 2 state in the hydrogen atom

(Given: $\frac{1}{R_H} = 912A°$, here R_H = Rydberg's constant)

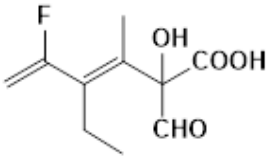
A) 434 B) 440 C) 490 D) 400

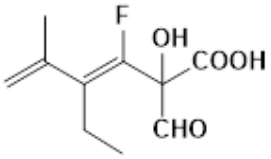
34. Which of the following is tertiary allyl alcohol

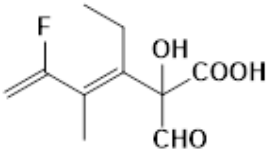
A)  B)  C)  D) 

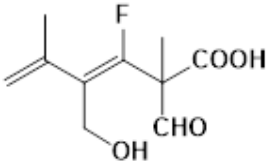
For More Material Join: @JEEAdvanced_2025

35. The structure of 3-ethyl-5-fluoro-2-formyl-2-hydroxy-4-methylhexa-3,5-dienoic acid

A) 

B) 

C) 

D) 

36. Match the list-I with list-II

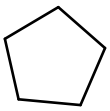
	List-I (molecule)		**List-II (shape)**
(A)	NH_3	(I)	Tetrahedral
(B)	IF_7	(II)	Trigonal bipyramidal
(C)	$\left[Ni(CO)_4\right]$	(III)	Pentagonal bipyramidal
(D)	PCl_5	(IV)	Pyramidal

The correct match is

A) $(A)-IV,(B)-II,(C)-I,(D)-III$ B) $(A)-IV,(B)-III,(C)-I,(D)-II$

C) $(A)-III,(B)-II,(C)-I,(D)-IV$ D) $(A)-III,(B)-I,(C)-II,(D)-IV$

37. The number of sp^2 hybridized carbon atoms present in the one molecule of final product A, formed in the following multistep reaction sequence is:



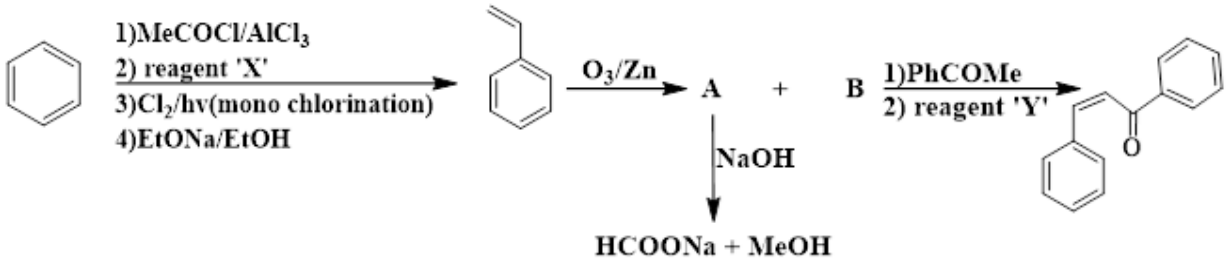
1)Cl_2/hv (mono chlorination)
2)Mg-dry ether
3)CO_2/NH_4Cl
4)NH_3/heat
5)Br_2/NaOH
6)PhCOCl /Py

→ **A**

A) 8 B) 5 C) 7 D) 6

For More Material Join: @JEEAdvanced_2025

38. In the given sequence of reactions identify the reagent 'X' and reagent 'Y'



A) X = NaOH , heat Y = N_2H_4 , NaOH

B) X = H_3O^+, heat Y = Zn-Hg + HCl

C) X = Zn-Hg + Conc.HCl Y = dil. *NaOH* , Heat

D) X = Pd – C / H_2 Y= $LiAlH_4$

39. Given below are two statements: One is labeled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A): $CH_2 = CHCl$ is an example of vinyl halide

Reason (R): Vinyl halides are the compounds in which the halogen atom is attached to sp hybridized carbon atom

In the light of the two above statements, choose the most appropriate answer from the options given below:

A) (A) is true but (R) is false

B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

C) (A) is false but (R) is true

D) Both (A) and (R) are true and (R) is the correct explanation of (A)

For More Material Join: @JEEAdvanced_2025

40. Which of the following statements is incorrect?

A) Two solutions having same osmotic pressure at a given temperature are called isotonic solutions

B) The solubility of gases increase with increase of pressure

C) According to Raoult's law, when a non-volatile solid is added to the solvent its vapour pressure decreases

D) Solubility of gases in liquids increases with rise in temperature

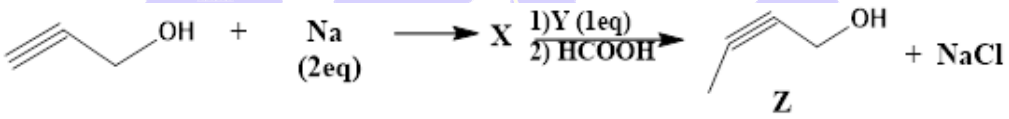
41. Match **List-I** with **List-2**

	List-I (element)		**List-II (E^0 of M^{+2}/M)**
(a)	Ti	(I)	-0.76V
(b)	Cr	(II)	+0.34 V
(c)	Cu	(III)	-1.63 V
(d)	Zn	(IV)	-0.90 V

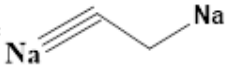
A) $(a)-III,(b)-IV,(c)-I,(d)-II$ B) $(a)-III,(b)-IV,(c)-II,(d)-I$

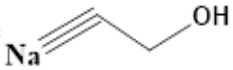
C) $(a)-III,(b)-II,(c)-I,(d)-IV$ D) $(a)-II,(b)-IV,(c)-I,(d)-III$

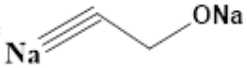
42. Compound X formed in the following reaction reacts with Y gives the product Z. Find out X and Y.



A) X=  Y= MeCl

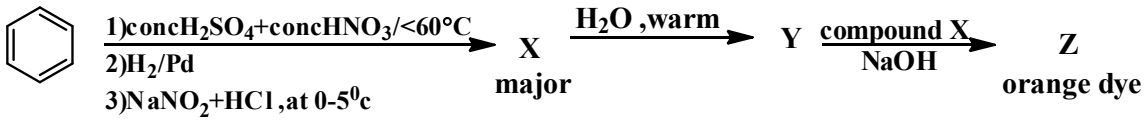
B) X=  Y= MeCl

C) X=  Y= MeCl

D) X=  Y= MeCl

For More Material Join: @JEEAdvanced_2025

43. The number of C=C bonds present in one molecule of final product Z, formed in the following multistep reaction sequence is:



A) 4 B) 5 C) 6 D) 7

44. Which of the following is correct about Lassaigne`s test?

A) The element nitrogen present in the organic compound is converted from covalent form into the ionic form KCN

B) The formation of Prussian blue colour confirms the presence of sulphur.

C) On treating sodium fusion extract with sodium nitroprusside, appearance of a violet colour further indicates the presence of nitrogen.

D) The sodium fusion extract is acidified with nitric acid and then treated with silver nitrate. A white precipitate, soluble in ammonium hydroxide shows the presence of chlorine

45. Choose the correct statements from the following

(I) $\left[Ag(NH_3)_2\right]\left[Ag(CN)_2\right]$ is named as: Diamminesilver (I) dicyanidoargentate (I)

(II) $\left[Ni(CO)_4\right]$ has tetrahedral geometry and diamagnetic

(III) $\left[Co(NH_3)_5(SO_4)\right]Br$ and $\left[Co(NH_3)_5Br\right]SO_4$ are ionization isomers

(IV) $\left[Co(en)_3\right]^{3+}$ has two optical isomers

(V) $\left[Co(NH_3)_6\right]^{3+}$ is a heteroleptic complex

A) (I),(II),(III) only B) (I),(III),(IV),(V) only

C) (I),(III),(IV) only D) (I),(II),(III),(IV) only

For More Material Join: @JEEAdvanced_2025

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

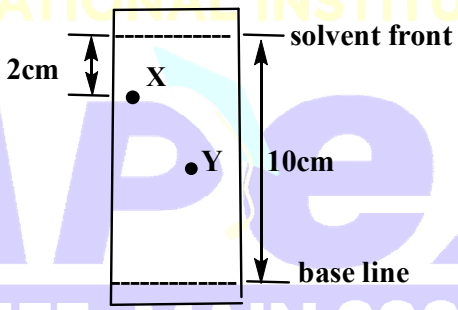
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The rate constants of a reaction at 500K and 700K are 0.02 s^{-1} and 0.07 s^{-1} respectively. Calculate the value of Ea in KJ $mole^{-1}$.
[log(0.07/0.02) = 0.544, R=8.3 J $K^{-1}mol^{-1}$]

47. The molar solubility of $Ni(OH)_2$ in 0.10 M NaOH is $Z\times10^{-13}$. The ionic product of $Ni(OH)_2$ is 2.0×10^{-15}, then value of 'Z' is

48. Calculate $\Delta_r G^0$ for conversion of oxygen to ozone, $\frac{3}{2}O_2(g)\rightleftharpoons O_3(g)$ at 298K. If K_p for this conversion is 2.47×10^{-29}. (report in kJ / mol); (Given: log2.47 = 0.392, R=8.314 JK^{-1} mol^{-1})

49. On a thin layer chromatographic plate, the retardation factor of the organic compound 'X' is $Z\times10^{-1}$,then value of 'Z' is



50. 13.8g of N_2O_4 was placed in a 1L reaction vessel at 400K and allowed to attain equilibrium

$$N_2O_4(g)\rightleftharpoons 2NO_2(g)$$

The total pressure at equilibrium was found to be 9.15 bar. The value of K_c at equilibrium is 'X', then value of 2X is ------ ($R = 0.083$ L.bar.K^{-1}.mol^{-1})

For More Material Join: @JEEAdvanced_2025

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. A line L_1; y = x + 2 is rotated through an angle of 90^0 in counter-clockwise direction about the point (0, 2) in xy plane and let new position of the line be $L_2 = 0$.Let L_3: y = -2x+10.The area of the triangle formed by lines $L_1 = 0, L_2 = 0$ and $L_3 = 0$ is equal to:

A) 56/3 B) 28/3 C) 64/3 D) 32/3

52. Remainder when $2025^{2024^{2023}}$ is divided by 7 is

A) 1 B) 4 C) 3 D) 2

53. Let $x,y \in R$ and $i = \sqrt{-1}$ satisfy the equation $(x+iy)^{2022} = x - iy$ .If the number of possible distinct ordered pairs (x, y) satisfying the given equation is N then the value of product of non zero digits of N is equal to:

A) 12 B) 4 C) 16 D) 8

54. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a}.\vec{b} = 0$. For some $x, y \in R$, let $\vec{c} = x\vec{a} + y\vec{b} + (\vec{a} \times \vec{b})$. If $|\vec{c}| = 2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $x^2 + y^2 + 8\cos^2\alpha$ is equal to:

A) 8 B) 4 C) 3 D) 6

55. The area of the largest rectangle having two vertices lying on the x-axis and the remaining vertices lying above the x-axis and on the parabola $y = 9 - x^2$ is equal to:

A) $12\sqrt{3}$ Sq.units B) $6\sqrt{3}$ Sq.units C) $24\sqrt{3}$ Sq.units D) $5\sqrt{3}$ Sq.units

56. Let $f(x) = \left(1 + \frac{1}{x}\right)^x, x > 0$.The value of $\lim_{n\to\infty}\left\{f\left(\frac{1}{n}\right).f\left(\frac{2}{n}\right).f\left(\frac{3}{n}\right).....f\left(\frac{n}{n}\right)\right\}^{1/n}$ is equal to:

A) $\sqrt{2}e$ B) $\sqrt{2e}$ C) $2\sqrt{e}$ D) $\sqrt{e}$

For More Material Join: @JEEAdvanced_2025

57. Let f(x) be a thrice differentiable function and satisfies $f\left(f\left(x\right)\right)=1-x$ for all $x\in\left[0,1\right]$

.If $J=\int_0^1 f\left(x\right)dx$ and $f''\left(\frac{4}{5}\right)=0$ then which of the following is NOT correct?

A) $f\left(\frac{1}{3}\right)+f\left(\frac{2}{3}\right)=1$

B) $J=\frac{1}{2}$

C) $f'''\left(x\right)=0$ has atleast one root in the interval $\left(\frac{1}{2},\frac{4}{5}\right)$.

D) $f'''\left(x\right)=0$ has no root in the interval $\left(\frac{1}{2},\frac{4}{5}\right)$.

58. If area formed by the set of points satisfying $-2\le y\le\sqrt{x-1}\le 2$ is

A) $\frac{40}{3}$ B) 40 C) $\frac{20}{3}$ D) $\frac{16}{3}$

59. Let y = y(x) be the solution of the differential equation

$\left(\cos ec\,x\right)dy+\left(2\left(1-x\right)\cot x+x\left(x-2\right)\right)dx=0$ such that $y\left(\frac{\pi}{2}\right)=3$ .The value of $y\left(2\right)$ is equal to:

A) 2 B) $2\left(1-\cos 2\right)$ C) $2\left(1+\cos 2\right)$ D) 3

60. If $\left(\alpha,\beta,\gamma\right)$ is the image of the point $\left(1,6,3\right)$ with respect to the line $\frac{x}{2}=\frac{y-1}{4}=\frac{z-2}{6}$ then the value of $\alpha^2+\beta^2+\gamma^2$ is equal to:

A) 26 B) 37 C) 17 D) 50

61. Two integers x and y are chosen one after the other with replacement from the set $\left\{0,1,2,3,.....,20\right\}$. Then the probability that $\left|x-y\right|>8$ is equal to:

A) $\frac{52}{147}$ B) $\frac{400}{441}$ C) $\frac{361}{441}$ D) $\frac{77}{147}$

For More Material Join: @JEEAdvanced_2025

62. Let $f(x)=\left[\log\left(\sin^{-1}\left(\sqrt{x^2+3x+2}\right)\right)\right]$ where $[.]$ is G.I.F. Which of the following is true?

A) Domain of definition of $f(x)$ is $\left[\frac{-3-\sqrt{5}}{2},\frac{-3+\sqrt{5}}{2}\right]$.

B) Domain of definition of $f(x)$ is $\left[\frac{-3+\sqrt{5}}{2},-1\right)$.

C) Range of $f(x)$ is the set of all non-positive integers.

D) Range of $f(x)$ is the set of all positive integers.

63. Consider the system of linear equations:

$2x+\left(p^2-2\right)y+6z=8, x+2y+(q-1)z=5, x+y+3z=4$ where $p,q\in R$. Which of the following statements is NOT correct?

A) The system will have unique solution if $p\in R-\{-2,2\}, q\in R-\{4\}$.

B) The system is inconsistent if $p=2,q=4$.

C) The system will have infinite solutions if $p=-2, q\in R$.

D) The system is consistent if $p\in R, q\in R-\{4\}$.

64. A line is touching two circles at T_1 and T_2 respectively such that the length T_1T_2 is 36. The minimum distance between the circles is 14. If the radius of the larger circle is four times the radius of the smaller circle then the circumference of the smaller circle is equal to:

A) 10π B) 8π C) 6π D) 20π

65. If the eccentricity of an ellipse $\frac{x^2}{\lambda^2+1}+\frac{y^2}{\lambda^2+2}=1$ is $\frac{1}{\sqrt{6}}$, then length of latus rectum of the ellipse is equal to:

A) $\frac{8}{\sqrt{6}}$ B) $\frac{10}{\sqrt{6}}$ C) $\frac{5}{\sqrt{6}}$ D) $\frac{7}{\sqrt{6}}$

For More Material Join: @JEEAdvanced_2025

66.

Class-interval	0-6	6-12	12-18	18-24	24-30
Frequency	4	5	3	6	2

The mean deviation from the median of the given data is equal to:

A) 7 B) 7.5 C) 6 D) 6.5

67. Let $A=\begin{bmatrix}1+\cos^2\theta & \sin^2\theta & 4\cos 6\theta \\ \cos^2\theta & 1+\sin^2\theta & 4\cos 6\theta \\ \cos^2\theta & \sin^2\theta & 1+4\cos 6\theta\end{bmatrix}$ and $f(\theta)=\det A$.The minimum value of $f(\theta)$for $\theta\in\left[\frac{\pi}{18},\frac{\pi}{9}\right]$is equal to:

A) -2 B) 0 C) 4 D) $2-2\sqrt{3}$

68. Let $\overrightarrow{AB}=3\vec{i}+4\vec{k}$ and $\overrightarrow{AC}=5\vec{i}-2\vec{j}+4\vec{k}$. If the length of altitude from A to the side BC of ΔABC is P then the greatest integer less than P^2 is equal to:

A) 20 B) 18 C) 19 D) 21

69. Let k be the number of real roots of the equation $4\sin^3 x+2\sin^2 x-2\sin x-1=0$ in the interval $[-\pi,\pi]$.If the roots of the equation $x^2-2ax+a^2+a-3=0$ are real and less than $\frac{k}{2}$then which of the following is true?

A) $a<2$ B) $2\le a\le 3$ C) $3<a\le 4$ D) $a>4$

70. Let $f:[0,1]\to R$ be a differentiable function such that $f(0)=0, f'(0)=12$.The value of $\lim\limits_{n\to\infty} n^2\int\limits_0^{1/n} f(t)\,dt$ is equal to:

A) 6 B) 12 C) 24 D) 0

For More Material Join: @JEEAdvanced_2025

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most 90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the maximum number of students who studied all the three subjects. The value of m + n is equal to

72. The value of $\int_{-1}^{2}\left[\frac{50x^2+[x]+50}{1+x^2}\right]dx$ (where [.] denotes greatest integer function) is equal to

73. For $x \in R$,Let $y=y(x)$ satisfies the differential equation $\left(x^2-5\right)dy+2x\left(\left(x^2-5\right)^2-y\right)dx=0$ such that $y(2)=7$.The value of $y(1)$ is equal to

74. Let A, G and H be the arithmetic mean, geometric mean and harmonic mean, respectively of two distinct positive real numbers. If α is the smallest of the two roots of the equation $A(G-H)x^2+G(H-A)x+H(A-G)=0$ then the value of $[\alpha+20]$ (where [.] denotes greatest integer function) is equal to:

75. Let $a=5^2+11^2+19^2+29^2+41^2+.....10terms$ and $b=\sum_{n=1}^{10}n^4$.The value of $\left[\frac{a-b}{909}\right]$ (where [.] denotes G.I.F) is equal to

For More Material Join: @JEEAdvanced_2025



For More Material Join: @JEEAdvanced_2025



For More Material Join: @JEEAdvanced_2025